

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations
 December 2023
MA1005-1 – MATRIX ALGEBRA AND DIFFERENTIAL EQUATIONS
 (For CV, ME, BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Rank of an identity matrix of order 4 is _____
 A) 1 B) 0
 C) 3 D) 4
- If eigen values of a 3×3 matrix A are $\lambda_1, \lambda_2, \lambda_3$ then the eigen values of A^{-1} are _____
 A) $-\lambda_1, -\lambda_2, -\lambda_3$ B) $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}$
 C) $\lambda_1^2, \lambda_2^2, \lambda_3^2$ D) $\lambda_1^{-2}, \lambda_2^{-2}, \lambda_3^{-2}$
- The value of k for which the following equations $x + 2y - 3z = 6, 2x - y + 4z = 2$ & $x + 2y + kz = 6$ has an infinite solution is _____
 A) -3 B) 3
 C) 2 D) 1
- If A is a singular matrix of order n then rank of A is _____
 A) n B) $n + 1$
 C) $< n$ D) $\leq n$
- The series $\frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots$, converges if _____
 A) $p > 1$ B) $p < 1$
 C) $p \leq 1$ D) $p \geq 1$
- If $\lim_{n \rightarrow \infty} u_n \neq 0$, then the series $\sum u_n$ must be _____
 A) Oscillatory B) Divergent
 C) Convergent D) Arithmetic series
- The n^{th} term of the series $\frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \dots$ is _____
 A) $\frac{2n+1}{n(n+1)(n+2)}$ B) $\frac{2n}{n(n-1)(n-2)}$
 C) $\frac{2n-1}{n(n-1)(n-2)}$ D) $\frac{2n-1}{n(n+1)(n+2)}$
- The Maclaurin's series expansion of e^x is _____
 A) $1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \infty$ B) $x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$
 C) $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$ D) $x - \frac{x^2}{2!} + \frac{x^3}{3!} - \dots \infty$

9.

The order and degree of the differential equation $\frac{d^2y}{dx^2} + 2\left(1 - \left(\frac{dy}{dx}\right)^3\right)^{\frac{1}{2}} = 0$ are _____

- A) 3, 2
C) 2, 1

- B) 2, 2
D) 3, $\frac{1}{2}$

10.

The differential equation $\frac{dy}{dx} + 5y = x^4y^7$ is _____

- A) Clairaut's equation
C) Linear equation

- B) Bernoulli's equation
D) Exact equation

11.

Integrating factor of the differential equation $\frac{dy}{dx} + \frac{1}{x \log x} y = \frac{2}{x}$ is _____

- A) $\frac{1}{\log x}$
C) $\frac{x^2}{e^{\frac{x}{2}}}$

- B) $e^{x \log x}$
D) $\log x$

12.

In order to find the differential equation of orthogonal trajectories of the curves $f(x, y, c) = 0$, which of the following replacements is made

- A) $\frac{dy}{dx}$ is replaced by $\frac{dx}{dy}$
C) $\frac{y}{x}$ is replaced by $\frac{x}{y}$

- B) $\frac{x}{y}$ is replaced by $-\frac{x}{y}$
D) $\frac{dy}{dx}$ is replaced by $-\frac{dx}{dy}$

13.

The complementary function of the differential equation $(D - 4)^2 y = e^{2x}$ is

- A) $y = (c_1 + c_2)x e^{4x}$
C) $y = (c_1 + c_2x) e^{-4x}$

- B) $y = (c_1 + c_2x) e^{4x}$
D) $y = c_1 e^{4x} + c_2 e^{-4x}$

14.

The number of arbitrary constants in the particular solution of a differential equation of fourth degree is

- A) 0
C) 3

- B) Same as its order
D) Insufficient data

15.

The particular integral of the differential equation $\frac{d^2y}{dx^2} = 2x$ is

- A) $y_p = \frac{x^2}{2}$
C) $y_p = \frac{x^3}{6}$

- B) $y_p = \frac{x^3}{3}$
D) $y_p = 1 + x$

16.

The value of W using the method variation of parameters for the differential equation $(D^2 + 4)y = e^{2x}$ is _____

- A) 1
C) e^{4x}

- B) 2
D) e^{2x}

17.

The value of $\int_0^{\infty} e^{-x^2} dx$ is _____

- A) $\sqrt{\pi}$
C) $\frac{\pi}{2}$

- B) $\sqrt{\frac{\pi}{2}}$
D) $\frac{\sqrt{\pi}}{2}$

18.

The value of $\int_0^{\frac{\pi}{2}} \sin^3 x \cos^4 x dx =$ _____

- A) $\frac{2}{35}$
C) $\frac{3\pi}{280}$

- B) $\frac{2}{3}$
D) $\frac{35}{2}$

19.

The value of $\int_0^1 \int_0^{\sqrt{x}} dy dx =$ _____

- A) $\frac{3}{2}$
C) $\frac{1}{3}$

- B) $\frac{2}{3}$
D) $\frac{9}{3}$

20.

The value of $\int_0^3 \int_0^2 \int_0^1 8xyz \, dx \, dy \, dz =$ _____

- A) 4
C) 36

- B) 9
D) 8

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Define the rank of a matrix. Find the rank of the given matrix using elementary row transformation.

$$\begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$$

6 L2 1 1

- b) Determine the largest eigen value and the corresponding eigen vector by the Rayleigh's power method for the matrix

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \text{ by taking } X^{(0)} = [1 \ 0 \ 0]^T. \text{ Carry out 5 iterations.}$$

6 L3 1 2

- c) Check for consistency of the given system of equations.

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 10 \\ 2x + 4y + 6z &= 8 \end{aligned}$$

4 L1 1 1

2. a) Find the eigen values and the eigen vectors of the matrix $\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$.
b) Use the Gauss-Seidel iteration method to Solve the following system of equations.

$$\begin{aligned} 2x + y - z &= 12 \\ 2x - 8y + 5z &= -5 \\ -5x + 3y + 9z &= -17, \text{ start with } x^{(0)} = y^{(0)} = z^{(0)} = 0 \text{ and carry out three iterations.} \end{aligned}$$

6 L2 1 1

- c) Using LU decomposition method, write the coefficient matrix of the given system as a product of lower and upper triangular matrices.

$$\begin{aligned} 2x + 3y + z &= 9 \\ x + 2y + 3z &= 6 \\ 3x + y + 2z &= 8 \end{aligned}$$

4 L1 1 1

Unit – II

3. a) Using the comparison test, check for convergence of the series $\frac{\sqrt{2}-1}{3^3-1} + \frac{\sqrt{3}-1}{4^3-1} + \frac{\sqrt{4}-1}{5^3-1} + \dots \infty$.
b) State the D'Alembert's ratio test and hence discuss for convergence of the series $\sum \frac{(n+3)!}{n! 3! 3^n}$.
c) Obtain the Maclaurin's series expansion of $\cos x$ up to three non-vanishing terms.

6 L2 2 1

6 L3 2 1

4 L1 2 2